

A4.3.7 Genetic Algorithms



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A4.3.7 describes genetic algorithms, applied in optimisation problems like route planning (travelling salesperson problem), using concepts like population, fitness, selection, crossover, mutation, evaluation, and termination

Introduction

We've seen how machines can learn from data and from interacting with an environment. Now, let's explore an approach inspired by the most powerful learning process on Earth: evolution.

Genetic Algorithms (GAs) are a type of search and optimisation technique that mimics the process of natural selection to find excellent solutions to complex problems.

What are Genetic Algorithms?

Genetic Algorithms are used to solve **optimisation problems**—problems where we need to find the best possible solution from a vast number of potential options.

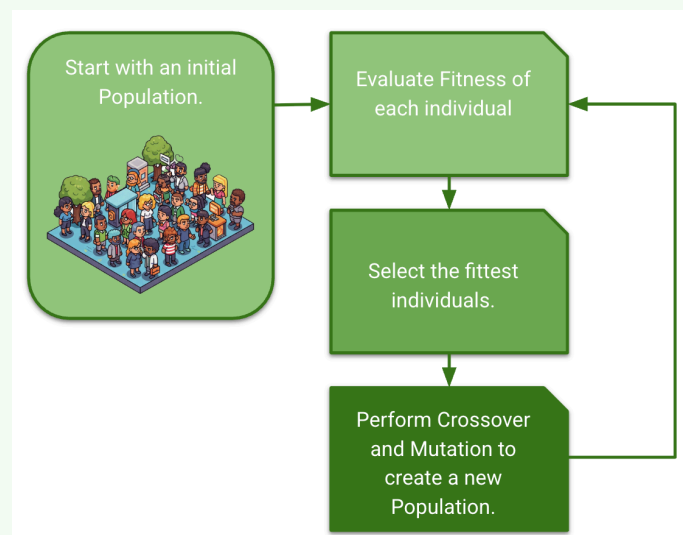
- **The Goal:** To "evolve" a solution. Instead of trying every single possibility (which would be impossible for complex problems), a GA starts with a set of random solutions and intelligently combines the best ones over many "generations" to find a near-optimal result.
- **Analogy:** It's like being a selective breeder for solutions. You start with a diverse group, only let the best ones "reproduce," and over time, the population gets better and better at the desired task.

The Steps of a Genetic Algorithm

A Genetic Algorithm follows an iterative process, like a cycle of life, death, and reproduction.

1. **Population:** The algorithm begins by creating an initial **population** of random candidate solutions. For the TSP, this would be a set of 100 completely random routes (random orders of visiting the cities).

2. **Fitness Function & Evaluation:** Each solution in the population is evaluated using a **fitness function**, which assigns it a score based on how good it is. For the TSP, a simple fitness function could be $1 / \text{total_distance}$. A shorter route has a higher fitness score.
3. **Selection:** The algorithm performs **selection**, where the fittest individuals from the population are chosen to be "parents" for the next generation. Solutions with higher fitness scores have a higher chance of being selected. This is "survival of the fittest."
4. **Crossover:** This is the "reproduction" step. The algorithm takes two parent solutions and performs **crossover** by combining parts of each to create a new "child" solution. For the TSP, this could mean taking the first half of Parent A's route and combining it with the second half of Parent B's route.
5. **Mutation:** To maintain genetic diversity and avoid getting stuck, the algorithm applies a **mutation** to the child solution. This is a small, random change. For the TSP, a mutation could be randomly swapping two cities in the child's route.
6. **Termination:** The new population of children replaces the old one, and the cycle (steps 2-5) repeats. This process is terminated when a specific condition is met, such as after a fixed number of generations or when the fitness of the best solution stops improving. The best solution found during the entire process is the final answer.



Case Study: The Travelling Salesperson Problem

To understand how a GA works, let's use a classic optimisation problem: the **Travelling Salesperson Problem (TSP)**.

The Problem: Given a list of cities, what is the shortest possible route that visits each city exactly once and returns to the starting city?

How can we find the best route without trying every single combination?

Click on the eye icon below to see how a GA can be applied to this problem.



- 1. Population:** Create a population of random solutions (e.g., 100 random routes for the salesperson).
- 2. Fitness Function:** Score every solution (e.g., $\text{fitness} = 1 / \text{total distance}$).
- 3. Selection:** Select the fittest solutions to be "parents."
- 4. Crossover:** Combine parts of two parent solutions to create a "child" solution.
- 5. Mutation:** Make a small, random change to a child solution.
- 6. Repeat:** The new children form the next generation. Repeat the cycle until the shortest route is found.

Flashcards

Use this Quizlet to learn all the key terms in this section.

Knowledge Check: Multiple Choice Quiz

Test your understanding of the core concepts.

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Genetic Algorithms are primarily used to solve what type of problem?

A. ☐ Regression

B. ☒ Optimization

C. ☐ Clustering

D. ☐ Classification

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The set of all candidate solutions in the current generation of a GA is known as the:

A. ☒ Population

B. ☐ Policy

C. ☐ Antecedent

D. ☐ Environment

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What is the purpose of the "fitness function" in a Genetic Algorithm?

A. ☐ To stop the algorithm when it is finished.

B. ☒ To evaluate how good a candidate solution is.

C. ☐ To create new solutions from parent solutions.

D. ☐ To introduce random changes into the population.

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The process of combining parts of two parent solutions to create a new child solution is called:

A. ☐ Evaluation

B. ☒ Crossover

C. ☐ Selection

D. ☐ Mutation

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In a GA for the Travelling Salesperson Problem, a "mutation" would most likely be:

- A. ☐ Combining two different routes.
- B. ☐ Deleting the worst-performing routes.
- C. ☐ Calculating the total distance of a route.
- D. ☒ Randomly swapping two cities in a single route.

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The "selection" process in a GA is inspired by which biological concept?

- A. ☒ Survival of the fittest
- B. ☐ DNA sequencing
- C. ☐ Asexual reproduction
- D. ☐ Random mutation

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Which of the following problems would be a good candidate for a Genetic Algorithm?

- A. ☐ Grouping customers into different market segments.
- B. ☐ Predicting the stock market price for tomorrow.
- C. ☒ Designing the most aerodynamic shape for a car to minimise drag.
- D. ☐ Classifying an email as spam or not spam.

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The condition that tells the Genetic Algorithm when to stop, such as after 1000 generations, is known as the:

A. ☐ Fitness function

B. ☒ Termination condition

C. ☐ Mutation rate

D. ☐ Selection criteria

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The purpose of mutation in a GA is to:

A. ☐ Score the solutions in the population.

B. ☒ Introduce genetic diversity and prevent the algorithm from getting stuck.

C. ☐ Combine the best parts of parent solutions.

D. ☐ Ensure the fittest individuals always survive.

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In a GA, the initial population of solutions is typically:

A. ☐ A set of known optimal solutions.

B. ☐ An empty set.

C. ☒ A set of randomly generated solutions.

D. ☐ A single, highly-fit solution.

